

Student Management System

ER Diagram Scenario

The edusphere Student Management System is designed to streamline the academic and administrative operations of ATI Tangalle. The system's database is structured around several interconnected entities that manage everything from user authentication to academic performance and student welfare.

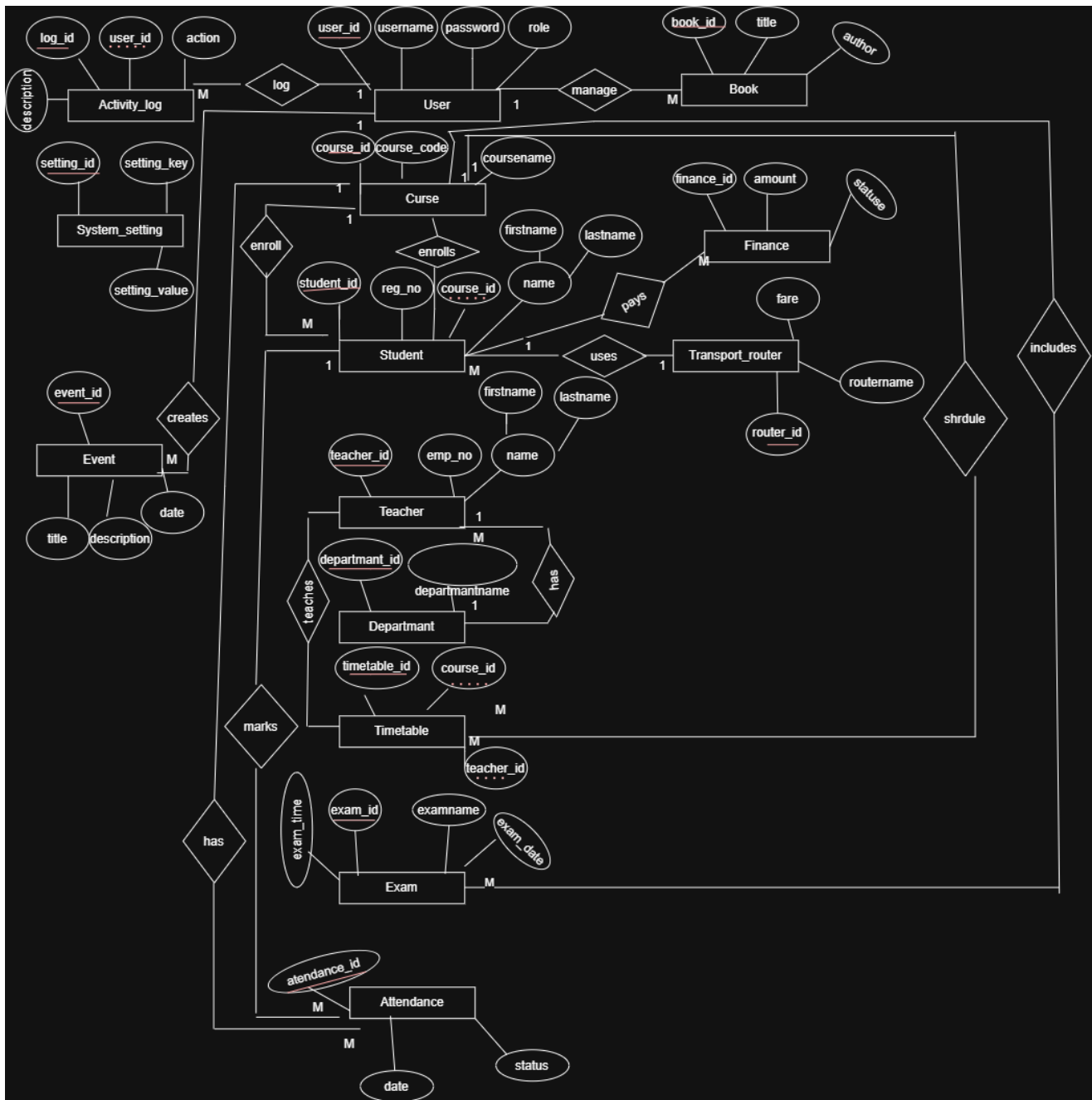
The process begins with the Administrator, who manages the Users (Admins, Teachers, and Students) and sets up the organizational structure by creating Departments and Courses. To maintain transparency and security, every action performed by the administrator is recorded in the Activity Log, and overall system preferences are maintained in the System Settings.

Once the setup is complete, Students enroll in specific Courses offered by different departments. The Teachers are assigned to these courses and are responsible for creating the Timetable, marking daily Attendance, and entering Exam marks. This creates a structured academic environment where every student's progress is tracked systematically.

Beyond academics, the system also manages student welfare and resources. Students can make payments through the Finance module, borrow books from the Books library, and utilize Transport Routes for commuting. Furthermore, the administrator can publish Events to keep the students and staff informed about upcoming activities or announcements.

In this scenario, all these entities are intricately linked—a student is connected to their courses, attendance, exams, and financial records, while teachers are linked to the courses they teach and the attendance they mark. This interconnected database ensures data consistency, eliminates redundancy, and provides a seamless, centralized platform for managing all aspects of the institute's daily operations.

ER Diagram



Use Case Diagram Scenario

The edusphere Student Management System is a comprehensive web-based application designed to streamline academic and administrative processes at ATI Tangalle. The system serves three primary actors: Admin, Teacher, and Student, each interacting with the system based on their specific roles and permissions.

1. Administrator (Admin) Role:

The Administrator acts as the primary controller of the system. After logging in, the Admin is responsible for setting up and managing the foundational data. This includes Managing Users (creating accounts for teachers and students), Managing Courses, and setting up the Timetable. The Admin also oversees resource management by handling Library operations, Hostel & Transport allocations, and student Finance records. Furthermore, the Admin can generate and view overall Reports and configure System Settings to ensure the system runs smoothly. When their tasks are complete, the Admin can securely log out.

2. Teacher Role:

Teachers are the academic facilitators within the system. Upon logging in, a Teacher can View Student Details to understand their class profile. Their core responsibilities include Marking Attendance for their assigned classes and Entering Exam Marks for students. Teachers can also View the Timetable to stay updated on their class schedules. Once their daily academic tasks are completed, they can log out of the system.

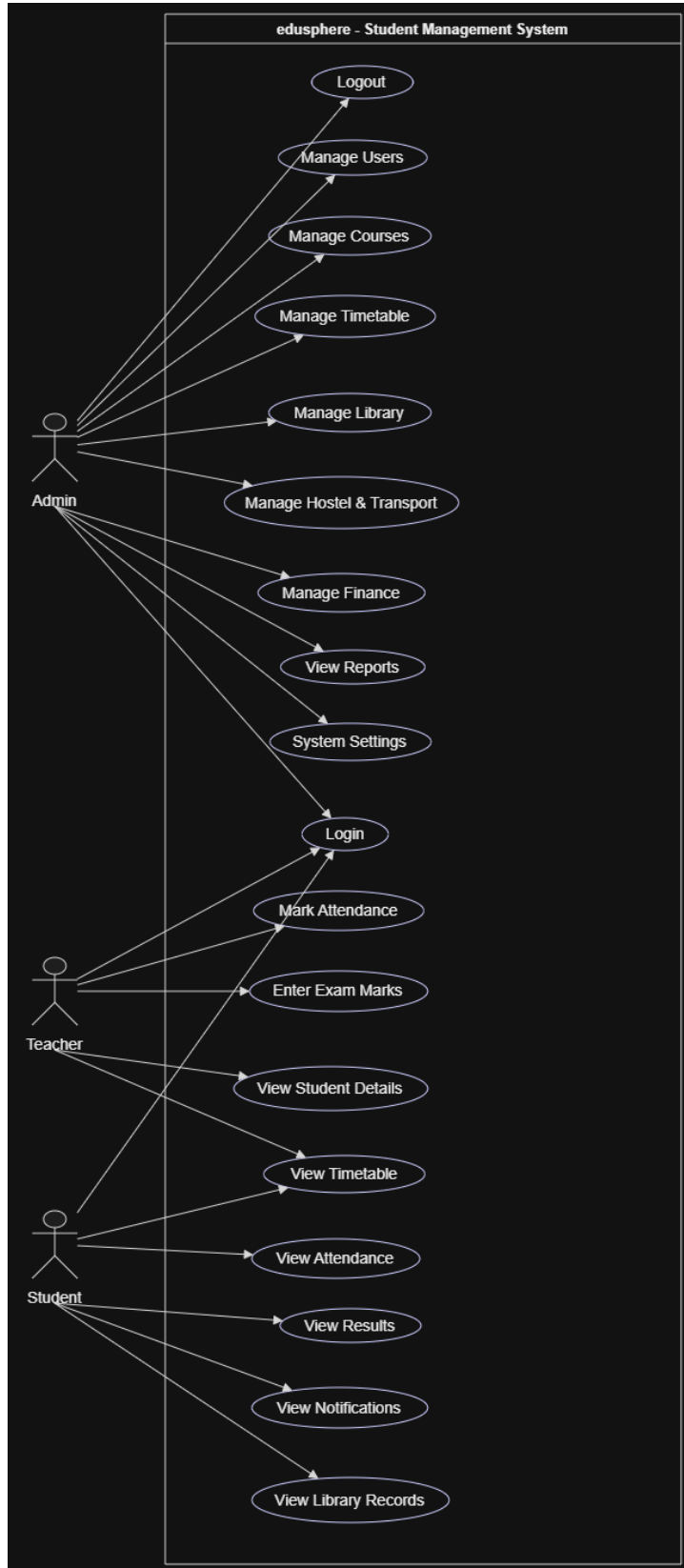
3. Student Role:

Students are the primary beneficiaries of the system. After logging in, a student can access a personalized dashboard to view their academic information. This includes Viewing Attendance records, checking Exam Results, and accessing the Timetable for their classes and exams. Students can also View Notifications regarding important events or announcements and check their Library Records for issued books. After reviewing their information, they can securely log out.

System Interaction:

All three actors interact with the system through a secure Login process, ensuring that data is protected and only accessible to authorized personnel. This clear separation of roles ensures that the edusphere system remains organized, secure, and efficient for all users.

Use Case Diagram



Level 0 (Context) Diagram Scenario

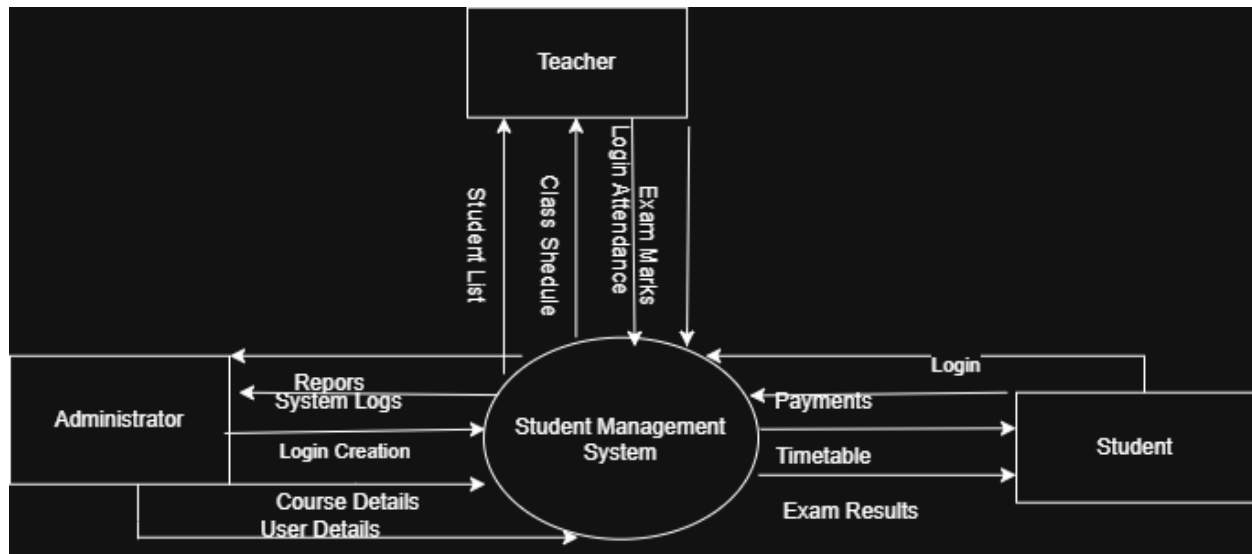
The Context Diagram (DFD Level 0) illustrates the highest-level view of the edusphere Student Management System, highlighting the interactions between the central system and its three primary external entities: Administrator, Teacher, and Student.

The Administrator acts as the primary controller of the system. The Admin inputs crucial data such as Login Creation credentials, User Details, and Course Details into the system. In return, the system generates Reports and System Logs, providing the Admin with the necessary oversight to manage the institute's operations.

The Teacher interacts with the system by providing a Login to securely access their dashboard. Teachers input daily Attendance records and Exam Marks for students. In return, the system provides them with Class Schedules and Student Lists, enabling them to efficiently manage their academic responsibilities.

The Student also logs into the system securely. Students provide Login credentials and make Payments for their academic or facility fees. The system processes these inputs and provides the students with their Timetable and Exam Results, allowing them to track their academic progress and schedules.

Level 0 (Context)Diagram



Level 1 DFD Diagram Scenario

The DFD Level 1 diagram refines the top-level context diagram by breaking down the edusphere Student Management System into four primary functional modules: Student Management, User Authentication, Academic Management, and Resource Management. This diagram illustrates how external entities interact with these specific processes and how data is stored within the system's database.

1. Process 1.0 – Student Management:

The Administrator initiates this process by providing User Details to the system. The process then stores this information in the D1: students data store. This module is responsible for managing the core profiles of the student body.

2. Process 2.0 – User Authentication:

This process handles security and access control for all users. The Administrator sends Login Creation data to this process. Meanwhile, both Teachers and Students submit their Login Credentials. The system validates these credentials against the D2: users data store. Upon successful validation, the system returns a Login Success message to the respective user, granting them access to their dashboards.

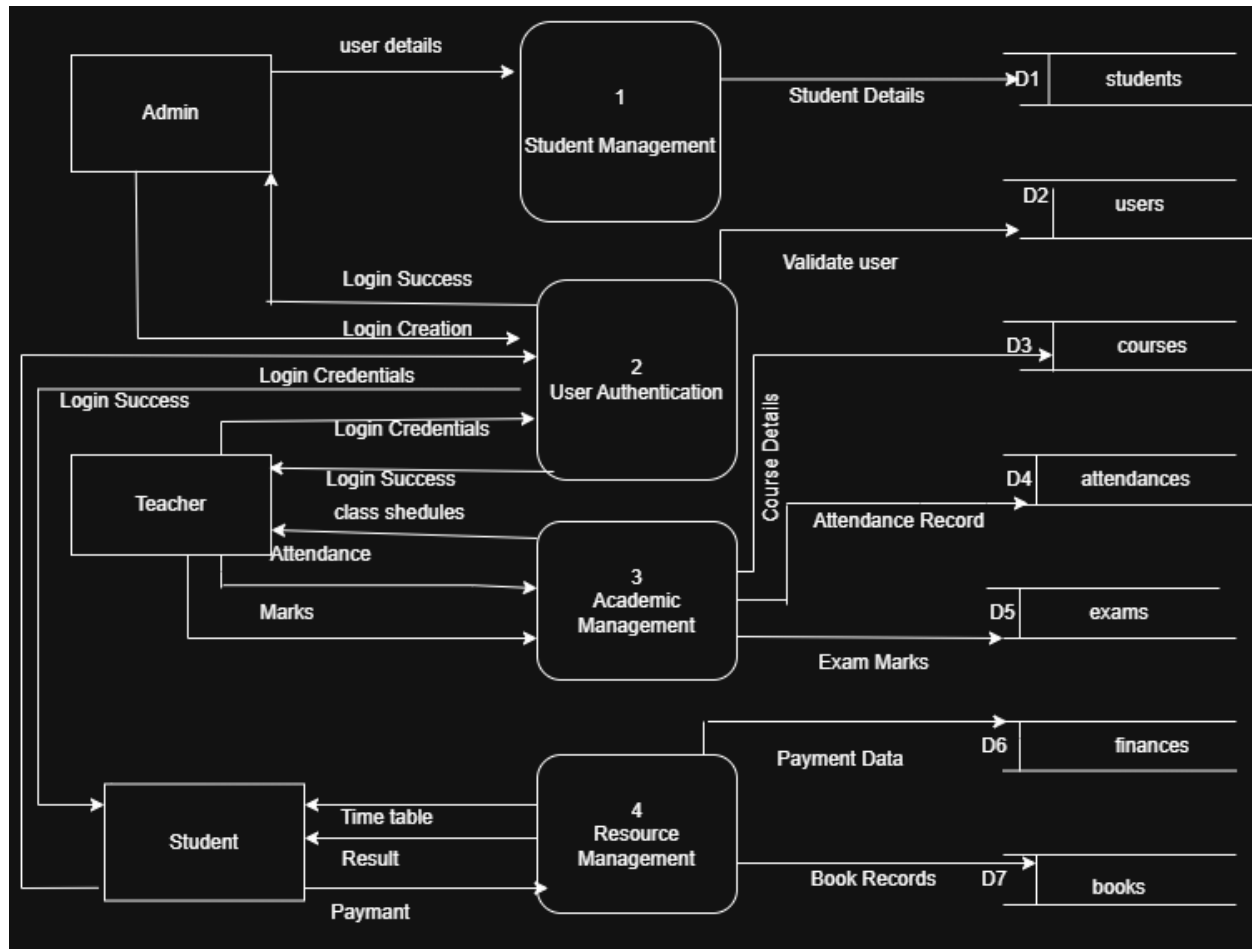
3. Process 3.0 – Academic Management:

This is the core academic hub. Teachers interact with this process by submitting Attendance records and Marks. The system processes this data and stores the Attendance Records in the D4: attendances data store and the Exam Marks in the D5: exams data store. Additionally, this process manages Course Details, updating the D3: courses data store, and provides the Teacher with their Class Schedules.

4. Process 4.0 – Resource Management:

This module handles student welfare and facilities. Students submit Payments to the system. The system processes this and stores the Payment Data in the D6: finances data store. The system also manages Book Records, updating the D7: books data store. In return for their inputs, the system provides students with their Timetable and Exam Results.

Level 1 DFD Diagram



Class Diagram Scenario

The Class Diagram of the edusphere Student Management System represents the static structure of the application, detailing the core classes, their attributes, methods, and the relationships between them. The system is anchored by a central User class, which acts as a base class for three distinct roles: Admin, Teacher, and Student. This inheritance structure allows all users to perform common actions like `login()`, `logout()`, and `change_Password()`, while having specialized functionalities based on their roles.

1. Academic Structure:

The Department class forms the top level of the academic hierarchy. Each department contains multiple Courses. These courses are linked to Teachers and Students, representing the core teaching and enrollment relationships. Teachers are responsible for marking Attendance and managing Exam marks, as well as scheduling Timetable entries for their classes.

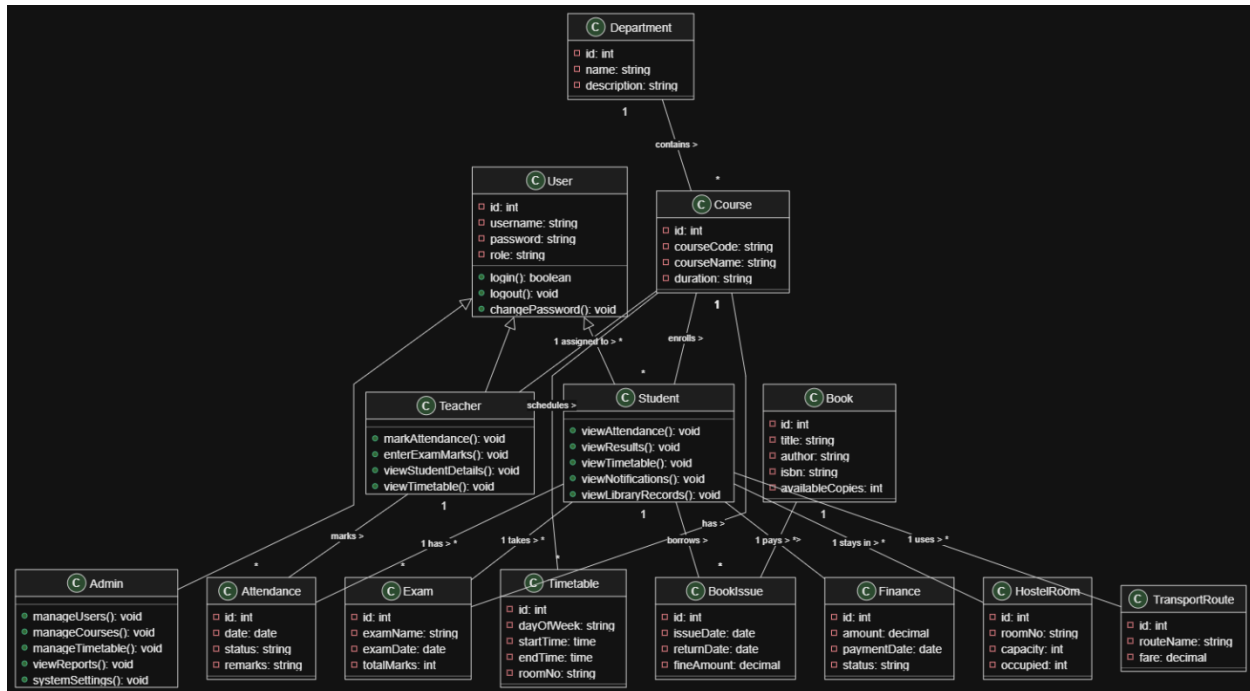
2. Student Services and Welfare:

Students can view their academic progress through methods like `viewAttendance()`, `viewResults()`, and `viewTimetable()`. Beyond academics, the system integrates student welfare services. Students can borrow Books (tracked via the BookIssue class), manage their Finance payments, and utilize HostelRoom and TransportRoute facilities. Each of these interactions is represented by a one-to-many relationship, indicating that a single student can have multiple financial records, book issues, or transport usages.

3. Administration:

The admin class acts as the system's superuser, possessing methods to `manageUsers()`, `manageCourses()`, and `manageTimetable()`. The admin is also responsible for `systemSettings()` and `viewReports()`, ensuring overall system governance and oversight.

Class Diagram



Sequence Diagram Scenario

The sequence diagram illustrates the step-by-step interaction between the user and the system during the login process. The primary participants in this scenario are the User (Admin, Teacher, or Student), the Login Page (User Interface), the Auth Controller (Business Logic), and the Database (users table).

Main Flow:

1. The process begins when the User enters their Username and Password on the Login Page and submits the form.
2. The Login Page captures these credentials and sends them to the Auth Controller for validation.
3. The Auth Controller queries the Database to find a user matching the provided username.
4. The Database returns the user data (if found) back to the Auth Controller.
5. The Auth Controller then performs a password check to verify if the entered password matches the stored one.

Alternative Flows (Alt Block):

Based on the password verification, the system splits into two possible outcomes:

- [Valid Credentials]: If the password matches, the Auth Controller returns a Login Success message to the Login Page. The Login Page then redirects the user to their respective Dashboard.
- [Invalid Credentials]: If the password does not match, the Auth Controller returns a Login Failed message. The Login Page then displays an Error Message to the user, prompting them to try again.

This sequence ensures a secure and systematic authentication process for all users accessing the edusphere system.

Sequence Diagram

